

Two players are playing a game of Tower Breakers! Player **1** always moves first, and both players always play optimally. The rules of the game are as follows:

- Initially there are n towers.
- Each tower is of height m .
- The players move in alternating turns.
- In each turn, a player can choose a tower of height x and reduce its height to y , where $1 \leq y < x$ and y evenly divides x .
- If the current player is unable to make a move, they lose the game.

Given the values of n and m , determine which player will win. If the first player wins, return **1**. Otherwise, return **2**.

Example. $n = 2$
 $m = 6$

There are **2** towers, each **6** units tall. Player **1** has a choice of two moves:

- remove **3** pieces from a tower to leave **3** as **6 modulo 3 = 0**
- remove **5** pieces to leave **1**

Let Player **1** remove **3**. Now the towers are **3** and **6** units tall.

Player **2** matches the move. Now the towers are both **3** units tall.

Now Player **1** has only one move.

Player **1** removes **2** pieces leaving **1**. Towers are **1** and **2** units tall.

Player **2** matches again. Towers are both **1** unit tall.

Player **1** has no move and loses. Return **2**.

Function Description

Complete the *towerBreakers* function in the editor below.

towerBreakers has the following parameter(s):

- int n*: the number of towers
- int m*: the height of each tower

Returns

- int*: the winner of the game

Input Format

The first line contains a single integer t , the number of test cases.

Each of the next t lines describes a test case in the form of **2** space-separated integers, n and m .

Constraints

- $1 \leq t \leq 100$
- $1 \leq n, m \leq 10^6$

Sample Input

STDIN	Function
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2	t = 2
2 2	n = 2, m = 2
1 4	n = 1, m = 4

Sample Output

2
1

Explanation

We'll refer to player **1** as **P1** and player **2** as **P2**

In the first test case, **P1** chooses one of the two towers and reduces it to **1**. Then **P2** reduces the remaining tower to a height of **1**. As both towers now have height **1**, **P1** cannot make a move so **P2** is the winner.

In the second test case, there is only one tower of height **4**. **P1** can reduce it to a height of either **1** or **2**. **P1** chooses **1** as both players always choose optimally. Because **P2** has no possible move, **P1** wins.